

Low-Light Image Enhancement with Cross-Dataset Generalization

Deep Learning Exam Project

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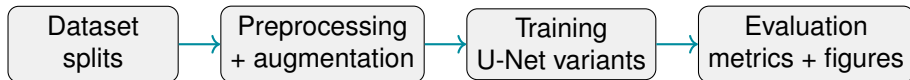
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Project Objective

Task: enhance low-light images while preserving structure, color balance, and cross-dataset robustness.

- ▶ Supervised training on **LOL-v2 Real Captured**.
- ▶ In-domain test on LOL-v2; cross-domain test on LOL-v1 and ExDark.
- ▶ Comparison between a compact baseline and controlled variants.
- ▶ Quantitative evaluation plus qualitative failure analysis.

Experimental Pipeline



Reproducibility

- ▶ fixed seed: 42;
- ▶ deterministic splits;
- ▶ saved checkpoints and histories;
- ▶ same protocol for all runs.

Simplicity

- ▶ compact PyTorch codebase;
- ▶ explicit dataset classes;
- ▶ no complex factories;
- ▶ readable exam-oriented design.

Dataset Protocol

Dataset	Role	Samples
LOL-v2 Real train	Training	586
LOL-v2 Real val	Validation split	103
LOL-v2 official test	In-domain paired test	100
LOL-v1 eval15	Cross-domain paired test	15
ExDark stratified	Cross-domain unpaired test	120

- ▶ LOL-v2 Real is associated with the semi-supervised LLIE work of Yang et al. [9].
- ▶ LOL-v1 originates from the RetinexNet dataset release [8].
- ▶ ExDark is used only for no-reference robustness analysis [2].

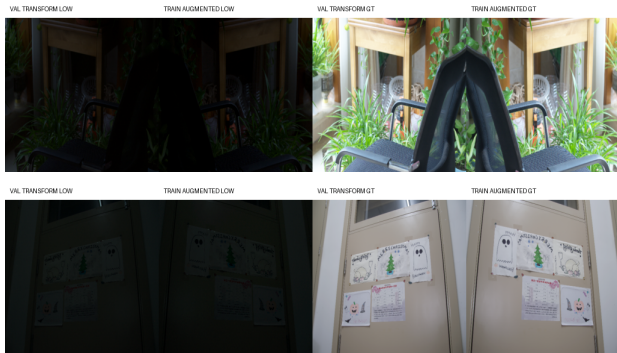
Preprocessing and Data Augmentation

Deterministic transforms

- ▶ RGB conversion;
- ▶ resize to 256×256 ;
- ▶ tensor conversion;
- ▶ normalization to $[0, 1]$.

Training augmentation

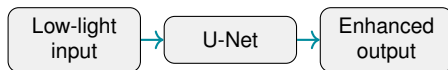
- ▶ random crop with padding;
- ▶ horizontal flip;
- ▶ mild brightness/contrast jitter on low image only.



Baseline: Compact U-Net

Design choice: classic encoder-decoder with skip connections, inspired by Ronneberger et al. [6].

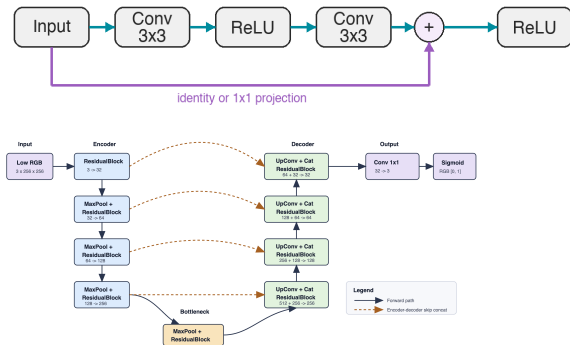
- ▶ double convolution blocks;
- ▶ max-pooling encoder;
- ▶ transposed-convolution decoder;
- ▶ skip concatenations;
- ▶ final sigmoid for $[0, 1]$ RGB output.



ResidualUNet Variant

Controlled architectural variant

- ▶ same high-level U-Net topology;
- ▶ DoubleConv blocks replaced by residual blocks;
- ▶ residual projection when channels change;
- ▶ no BatchNorm, to preserve intensity and color statistics.



Loss Functions

Baseline reconstruction objective

$$\mathcal{L}_{base} = 0.8\mathcal{L}_{L1} + 0.3\mathcal{L}_{SSIM}$$

L1 reconstruction

$$\mathcal{L}_{L1} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|$$

Pixel-level fidelity.

SSIM loss

$$\mathcal{L}_{SSIM} = 1 - \text{SSIM}(\hat{y}, y)$$

Local luminance, contrast and structure [7].

The same loss is reused for the baseline U-Net and ResidualUNet.

Color Constancy Mitigation

Observed failure: the baseline can brighten images while introducing grayish or desaturated colors.

$$\mathcal{L}_{color} = \frac{1}{B} \sum_{i=1}^B \|\mu_{RGB}(\hat{y}_i) - \mu_{RGB}(y_i)\|_1$$

- ▶ Inspired by color constancy and Gray-World assumptions [1].
- ▶ Supervised version: match RGB channel means of the ground truth.
- ▶ Added as a weak regularizer: $\lambda_{color} = 0.05$.

Important: this is not a new architecture; it isolates the effect of a color-aware objective.

Training Setup

Configuration

- ▶ optimizer: AdamW [3];
- ▶ learning rate: $3 \cdot 10^{-4}$;
- ▶ batch size: 8;
- ▶ weight decay: 10^{-5} ;
- ▶ max epochs: 100;
- ▶ early stopping patience: 10.

Hardware

- ▶ MacBook Air M4;
- ▶ 24 GB unified memory;
- ▶ MPS backend when available;
- ▶ mixed precision disabled on MPS.

Checkpoint selection

Early stopping is based on validation loss, because it matches the optimized objective.

Evaluation Metrics

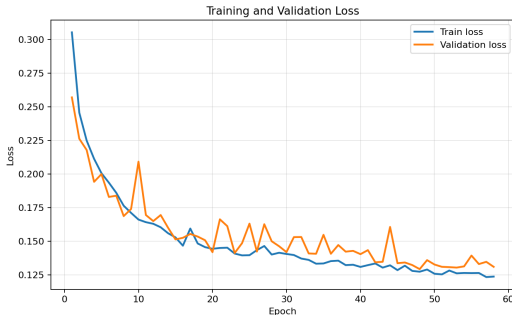
Metric	Meaning	Used for
PSNR $\uparrow$	MSE-based fidelity	paired datasets
SSIM $\uparrow$	local structural similarity	paired datasets
NIQE $\downarrow$	blind naturalness score	paired + unpaired
BRISQUE $\downarrow$	blind spatial quality score	ExDark only

- ▶ PSNR and SSIM are primary metrics when ground truth is available.
- ▶ NIQE and BRISQUE are interpreted carefully: they do not measure task fidelity [5, 4].
- ▶ For NIQE, $\Delta < 0$ means the prediction improves over the low-light input.

Validation Dynamics

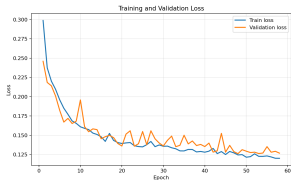
Run	Epochs	Loss	PSNR
Baseline	58	0.1262	19.775
ResidualUNet	36	0.1324	19.434
Color Loss	58	0.1293	19.761

- ▶ all runs converge before 100 epochs;
- ▶ color term does not destabilize training;
- ▶ ResidualUNet stops earlier.

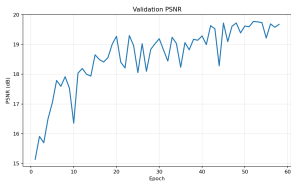


Training Curves

Baseline

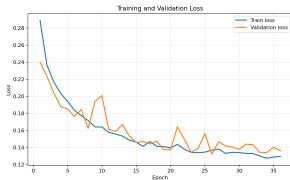


Train/val loss

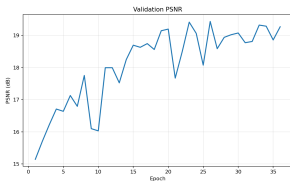


Validation PSNR

ResidualUNet

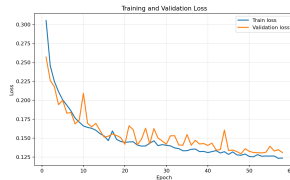


Train/val loss

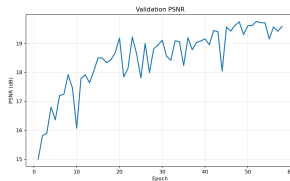


Validation PSNR

Color Loss



Train/val loss



Validation PSNR

Paired Test Results

Dataset	Run	PSNR	SSIM	NIQE
LOL-v2	Baseline	17.726	0.8359	8.384
	ResidualUNet	18.251	0.8355	8.180
	Color Loss	17.850	0.8362	7.316
LOL-v1	Baseline	20.031	0.8361	8.430
	ResidualUNet	18.509	0.8197	7.414
	Color Loss	19.994	0.8373	6.924

Interpretation

ResidualUNet improves in-domain PSNR, while the color-loss baseline gives the best NIQE with almost unchanged PSNR/SSIM.

ExDark: Unpaired Robustness

Run	NIQE	BRISQUE
Input	8.116	31.055
Baseline	7.773	39.947
ResidualUNet	8.302	41.985
Color Loss	7.113	39.154

- ▶ ExDark has no ground truth.
- ▶ NIQE improves most with color loss.
- ▶ BRISQUE worsens vs input: enhancement can amplify artifacts.
- ▶ The subset is for robustness and qualitative analysis, not a full benchmark.

Qualitative Example: LOL-v2

LOW INPUT

Baseline

Baseline Color Loss

Residual Unet

GROUND TRUTH



The color-loss model reduces the grayish cast of the baseline while preserving the global enhancement behavior.

Qualitative Example: Cross-Dataset



LOL-v1 paired evaluation



ExDark unpaired evaluation

Main observation

Cross-domain examples expose differences in camera pipeline, exposure, noise, and scene content.

Failure Taxonomy

Failure	Visible effect	Mitigation
Color cast	output becomes gray-green or desaturated	color constancy loss*
Halo artifacts	bright bands around local light sources	gradient-aware loss
Over-smoothing	thin structures and textures become soft	edge/perceptual loss
Hallucinated details	distorted contours in very dark regions	confidence-aware enhancement
Residual noise	dark background remains noisy or patchy	noise-aware training

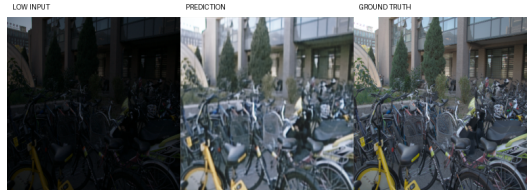
*Only color cast was tested; the other mitigations are future experiments.

Failure Examples: Color and Smoothing



Color cast

Examples are taken from baseline U-Net predictions.



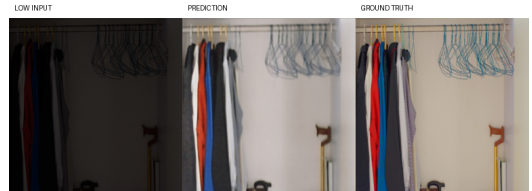
Over-smoothing

Failure Examples: Halo and Hallucination



Halo artifacts

Examples are taken from baseline U-Net predictions.



Hallucinated details

Failure Example: Residual Noise

LOW INPUT



PREDICTION



Residual noise and patchy dark background.

Main Takeaways

1. **The compact U-Net is a useful baseline.**
 - ▶ It enhances visibility and provides a transparent reference.
2. **More capacity does not guarantee robustness.**
 - ▶ ResidualUNet improves LOL-v2 PSNR but transfers worse.
3. **Color statistics are a key cross-domain weakness.**
 - ▶ A small color-constancy term improves no-reference scores and visual color balance.

Conclusion

The strongest improvement is not architectural, but loss-level: explicitly regularizing global color balance makes the baseline more visually reliable.

Future Work

- ▶ **Complete the mitigation study.**
 - ▶ Test gradient-aware losses for halo artifacts.
 - ▶ Add edge/perceptual constraints for over-smoothing.
 - ▶ Explore noise-aware training for residual noise.
- ▶ **Apply the same mitigations to ResidualUNet.**
 - ▶ This would separate architecture effects from loss-function effects.
- ▶ **Keep the protocol cross-dataset.**
 - ▶ Improvements should be checked on LOL-v2, LOL-v1, and ExDark.

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- [6] Olaf Ronneberger, Philipp Fischer e Thomas Brox. «U-Net: Convolutional Networks for Biomedical Image Segmentation». In: *MICCAI*. 2015.
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